

Comprehensive Comparison: DICE-2013 vs DICE-2023

Major Structural Changes

Component	DICE-2013	DICE-2023	Change Impact
Version Name	DICE-2013R	DICE-2023	Updated January 2024 knowledge cutoff
Base Year	2010	2020	10-year update
Last Prior Version	DICE-2007	DICE-2016	Different update cycles

1. Carbon Cycle and Climate System

Feature	DICE-2013	DICE-2023	Significance
Carbon Cycle Model	Linear three-reservoir model	DFAIR (DICE-FAIR) model - nonlinear four-reservoir impulse-response	Major change: Accounts for declining sink capacity with higher emissions
Carbon Cycle Structure	Linear absorption rates	Nonlinear with saturation parameter $\alpha(t)$ that increases atmospheric retention as cumulative emissions rise	More realistic: 70% retention at 5,000 GtC vs 30% at 100 GtC pulse
Carbon Reservoirs	3 reservoirs	4 reservoirs (R_1, R_2, R_3, R_4) with time constants τ_i	Based on FAIR model (Millar et al. 2017)
Atmospheric Retention	Fixed linear rates	Dynamic: $iIRF100(t) = \xi_0 + \xi C \cdot Cacc(t) + \xi T \cdot TAT(t)$	Temperature and cumulative emission dependent
Climate Model	Simple two-box model	Two-box model updated with AR6 parameters	Updated parameters
Equilibrium Climate Sensitivity (ECS)	$\sim 3.0^{\circ}C$	$3.0^{\circ}C$ (centered on IPCC AR6)	Maintained but recalibrated
Transient Climate Response (TCR)	Not explicitly stated	$1.8^{\circ}C$	Explicitly calibrated to IPCC AR6
Climate Science Base	IPCC AR4/AR5	IPCC AR6 (2023) and FAIR model	8+ years of new science

2. Damage Function

Feature	DICE-2013	DICE-2023	Impact
Functional Form	Quadratic: $\Omega(t) = \psi_1 \cdot TAT(t) + \psi_2 \cdot [TAT(t)]^2$	Same quadratic form	Maintained
Damage at 3°C	$\sim 1.2\%$ of GDP	3.1% of GDP	2.6x increase

Feature	DICE-2013	DICE-2023	Impact
Damage at 4.5°C	Not specified	7.0% of GDP	Higher damages
Literature Base	Earlier synthesis	56 estimates from 33 studies (updated through 2023)	Expanded evidence base
Tipping Points	Not explicitly included	+1% GDP loss at 3°C (Dietz et al. 2021)	New component
Omitted Impacts Adjustment	Implicit	+0.5% GDP loss at 3°C (judgmental)	Explicit adjustment
Components	Single estimate	Three components: (1) literature synthesis, (2) tipping points, (3) omitted impacts	More comprehensive
Damage Coefficient	$\psi_2 \approx 0.00236$	$\psi_2 \approx \mathbf{0.00458}$	~94% increase
Calibration Range	Up to ~3°C	Up to 4°C (quadratic not valid beyond)	Better specified limits

3. Greenhouse Gas Emissions

Feature	DICE-2013	DICE-2023	Significance
Controllable Emissions	Industrial CO ₂ only	All abatable emissions (industrial CO ₂ + land-use CO ₂ + abatable non-CO ₂ GHGs)	Major expansion
CO ₂ -equivalent Coverage	Industrial CO ₂	140% of industrial CO ₂ in 2020, declining to 121% in 2100	Much more comprehensive
Non-CO ₂ GHGs	Exogenous forcings	Endogenous abatable non-CO₂ with control rate $\mu(t)$	Policy-responsive
Non-CO ₂ Baseline	Fixed external forcing	SSP2 scenario baseline with abatement costs	Dynamic
Control Rate Application	CO ₂ only	Same $\mu(t)$ for CO ₂ and non-CO ₂ abatable emissions	Unified framework
Land-use Emissions	Exogenous	Included in total abatable CO ₂ (declining 2%/year)	More complete
Emissions-Output Ratio	$\sigma(t)$, declining 1.5%/year	$\sigma(t)$, declining 1.5%/year initially	Similar trend

4. Abatement Costs and Backstop Technology

Feature	DICE-2013	DICE-2023	Change
Cost Function	$\Lambda(t) = \theta_1(t) \cdot \mu(t)^{\theta_2}$	$\mathbf{\Lambda(t) = \theta_1(t) \cdot \mu(t)^{\theta_2}}$ (scaled for all GHGs)	Expanded scope
Backstop Price (2050)	Lower (~\$300/tCO ₂ , estimated)	\$515/tCO₂ (2019\$, median from ENGAGE study)	Higher cost

Feature	DICE-2013	DICE-2023	Change
Backstop Price Basis	Process models	Multi-model comparison (ENGAGE, Riahi et al. 2021)	Empirical basis
Decline Rate (2020-2050)	~0.5%/year	1.0%/year	Faster cost reduction
Decline Rate (post-2050)	Not specified	0.1%/year	Slower long-term decline
Zero Emissions Cost (2020)	Not specified	11% of GDP	Specified
Zero Emissions Cost (2100)	Not specified	2.7% of GDP	Declining costs
Non-CO₂ Abatement	Not modeled	Separate cost function based on MAC curves	New feature

5. Discounting and Risk

Feature	DICE-2013	DICE-2023	Major Innovation
Discount Rate Approach	Simple time preference ρ	Risk-adjusted certainty equivalent	Fundamental change
Pure Rate of Time Preference (ρ)	1.5%/year	1.5%/year base	Same base
<i>Risk-Adjusted Rate $\rho(t)^*$</i>	Not applicable	$\rho(t) = \rho - \frac{1}{2}\phi^2\sigma^2 \cdot t + \beta CLIM \cdot \pi^*$	New framework
Precautionary Term	Not included	$-\frac{1}{2}\phi^2\sigma^2 \cdot t$ for consumption growth uncertainty	New
Climate Beta Adjustment	Not applicable	$\beta CLIM = 0.5$ (climate investment risk)	New concept
Growth Uncertainty (σ)	Not modeled	1%-point/year standard deviation	Explicit
Risk Premium (π)	Not applicable	5%/year economy-wide	Market-based
Near-term Discount Rate	~4%/year	4.5%/year initially , declining over time	Dynamic
Long-term Behavior	Constant	Declining with time	Time-varying
Consumption Elasticity (ϕ)	1.45	1.45	Unchanged

6. Economic Parameters

Feature	DICE-2013	DICE-2023	Change
Production Function	Cobb-Douglas	Cobb-Douglas	Maintained

Feature	DICE-2013	DICE-2023	Change
Capital Share (γ)	0.30	0.30 (typical)	Standard
Population	UN 2010 projections	UN projections (updated)	New base
TFP Growth	Historical extrapolation	Based on Christensen et al. (2018), Müller et al. (2022), Newell et al. (2022), Rennert et al. (2021)	Updated sources
Output Measurement	PPP exchange rates	PPP exchange rates (World Bank, IMF)	Same method
Time Step	10 years initially, 5 years later	5 years throughout	More granular
Time Horizon	To 2595	To ~2150-2300	Similar long-term

7. Policy Scenarios

Scenario	DICE-2013	DICE-2023	Notable Differences
Baseline	Business-as-usual	Current policy (\$6/tCO ₂ , growing 2.5%/year)	More realistic baseline
Optimal	Cost-benefit optimal	"C/B Optimal" (renamed for clarity)	Terminology change
Temperature-Limited	2°C and 2.5°C limits	2°C limit (1.5°C infeasible)	Focused targets
Paris Accord	Not included	Extended Paris scenario (pledges through 2030, extrapolated)	New scenario
Alternative Discounting	Limited analysis	Systematic 1-5% constant rate scenarios	Expanded sensitivity
Alternative Damages	Not systematic	Howard-Stern damage function (9% at 3°C)	Explicit alternative

8. Key Results Comparison

Metric	DICE-2013	DICE-2023	Change Direction
Social Cost of Carbon (2020)	~44/tCO ₂ (2015, optimal)	66/tCO ₂ (2019, baseline)	~50% increase
SCC (C/B optimal, 2020)	Lower	\$50/tCO ₂	Comparable accounting for inflation
SCC (2°C limit, 2020)	Not specified	\$76/tCO ₂	New result
Optimal Temperature (2100)	~3.5°C	2.6°C	0.9°C lower

Metric	DICE-2013	DICE-2023	Change Direction
Baseline Temperature (2100)	~4.1°C	3.6°C	Lower due to current policies
Paris Temperature (2100)	Not modeled	~2.8°C	New pathway
Optimal Emissions Control (2050)	Lower	39%	Higher early action
2°C Emissions Control (2050)	Higher	55%	More stringent
Cost of 2°C Limit	Higher relative to optimal	Lower cost (closer to optimal)	More achievable
Wealth Gain (C/B optimal)	Not specified	\$120 trillion PV	Substantial
Emissions Path	More gradual	Steeper early reductions	Urgency increased

9. Model Output Variables

Variable	DICE-2013	DICE-2023	Expansion
GDP	✓	✓	Same
Damages	✓	✓	Same
Abatement Costs	✓	✓	Expanded to all GHGs
Industrial CO ₂ Emissions	✓	✓	Same
Land-use CO ₂ Emissions	Exogenous	✓	Now controllable
Abatable Non-CO ₂ Emissions	Not tracked	✓ (new)	Major addition
Carbon Prices	✓	✓	Same
Temperature Change	✓	✓	Same
Atmospheric Concentrations	✓	✓	More accurate
Consumption	✓	✓	Same
Investment	✓	✓	Same
Social Cost of Carbon	✓	✓	Enhanced methodology

10. Implementation Constraints

Feature	DICE-2013	DICE-2023	Difference
Maximum Control Rate Increase	Not specified	12 percentage points per 5 years	Implementation limit
Maximum Control Rate (through 2120)	100%	100%	Same

Feature	DICE-2013	DICE-2023	Difference
Maximum Control Rate (post-2120)	Not applicable	110 % (negative emissions)	New feature
Participation	Assumed	100 % country participation assumed	Explicitly stated
Carbon Pricing	Varied	Harmonized and comprehensive assumed	Idealized
1.5°C Target	Not evaluated	Declared infeasible within model constraints	Important finding

11. Software and Computational

Feature	DICE-2013	DICE-2023	Update
Primary Platform	GAMS	GAMS	Same
Excel Version	Available (simplified)	Available (with caveats)	Maintained
MATLAB Version	Not official	Available (community, 2016 version)	Community contribution
Documentation	User manual	User manual + PNAS paper + appendices	Enhanced
Code Availability	dicemodel.net	https://bit.ly/3TwJ5nO	Public
Solution Method	Nonlinear programming	Nonlinear programming	Same

12. Scientific Updates

Area	DICE-2013	DICE-2023	Advancement
IPCC Report Base	AR4/AR5 (2007/2013)	AR6 (2021-2023)	Latest science
Carbon Cycle Science	Traditional linear	FAIR model (Millar et al. 2017)	State-of-art
Damage Studies	Through ~2013	Through 2023 (56 estimates, 33 studies)	Expanded literature
Tipping Points	Qualitative concern	Quantified (Dietz et al. 2021: +1% at 3°C)	Explicit inclusion
Risk Framework	Standard utility	Climate beta (Dietz et al. 2018)	Financial theory integration
Backstop Technology	Theoretical	Empirical (ENGAGE multi-model)	Grounded estimates

13. Policy Implications

Finding	DICE-2013	DICE-2023	Implication
Optimal vs 2°C Gap	Large gap	Smaller gap (2.6°C vs 2°C optimal)	2°C more justified

Finding	DICE-2013	DICE-2023	Implication
Current Policies	Inadequate	Grossly inadequate (3.6°C by 2100)	Greater urgency
Paris Accord Impact	Not evaluated	Reduces 1/3 of gap from baseline to 2°C	Helpful but insufficient
Near-term Action	Gradual ramp	Steeper early reductions needed	Front-loaded effort
SCC for Policy	~\$44	\$66-76 depending on scenario	Higher carbon pricing justified
Economic Stakes	Significant	\$120 trillion wealth gain from optimal policy	Enormous benefits

14. Caveats and Limitations (Explicitly Noted)

Limitation	DICE-2013	DICE-2023	Evolution
Damage Function Uncertainty	Noted	Heavily emphasized (limited evidence >4°C)	More explicit
Quadratic Limitations	Implicit	Explicit warning about threshold damages	Better specified
Implementation Challenges	Acknowledged	100% participation assumption is "highly optimistic"	More realistic
Technology Assumptions	Fixed	Exogenous and fixed explicitly noted as limitation	Transparent
Short-run Dynamics	Not designed for	5-year timestep limits short-run analysis	Explicit scope
Distributional Effects	Aggregate	Representative agent - no within-country distribution	Clear limitation
Geoengineering	Not included	Solar geoengineering not considered	Explicit exclusion

Summary of Most Important Changes

Top 5 Structural Innovations in DICE-2023:

1. **DFAIR Nonlinear Carbon Cycle** - Captures declining sink capacity, more realistic atmospheric retention at high emissions
2. **Comprehensive GHG Coverage** - All abatable emissions (CO₂ + non-CO₂) now policy-responsive
3. **Risk-Adjusted Discounting** - Climate beta and consumption growth uncertainty incorporated
4. **Higher Damage Function** - ~2.6x higher (3.1% vs 1.2% at 3°C), with tipping points explicit
5. **Empirical Backstop Costs** - Based on multi-model comparison rather than theoretical estimates

Top 5 Results Changes:

1. **Lower Optimal Temperature** - 2.6°C vs 3.5°C in 2100

2. **Higher Social Cost of Carbon** - \$66 vs ~\$44 (real increase ~30-40%)
3. **2°C Target More Justified** - Closer to cost-benefit optimal due to higher damages
4. **Paris Accord Quantified** - Achieves 1/3 of needed reductions from baseline to 2°C
5. **1.5°C Declared Infeasible** - Would require immediate 50%+ emissions cuts or deep recession

Key Philosophy Changes:

- **More Precautionary:** Higher damages, explicit tipping points, risk-adjusted discounting
- **More Comprehensive:** All GHGs, not just industrial CO₂
- **More Realistic:** Implementation constraints, realistic baseline policies
- **More Transparent:** Explicit limitations, alternative scenarios, sensitivity analysis
- **More Policy-Relevant:** Paris Accord evaluation, wider range of targets

This comparison is based on DICE-2013R User Manual (Nordhaus & Sztorc 2013), DICE-2023 materials (Barrage & Nordhaus 2024 PNAS paper), and supporting documentation from June 2025.